

Yingrui Hou

Applied Data Scientist | Statistical Modelling, Validation & Monitoring

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SUMMARY

Applied data scientist with a background in experimental physics and several years of experience building Python/C++ workflows for calibration, validation, inference, and monitoring on noisy real-world measurement systems.

Experience spans calibration, uncertainty-aware analysis, weak-label inference, dataset-shift mitigation, anomaly diagnostics, and reproducible pipelines for complex measurement data.

Best fit: applied data science roles in analytics, model validation, monitoring, anomaly detection, or decision support for complex measurement data.

CORE SKILLS

- Core data science stack: Python, Statistical Modelling, Model Validation, Calibration, Monitoring
- Libraries and ML: scikit-learn, pandas, NumPy, statsmodels, CatBoost / Gradient Boosting
- Applied methods: Anomaly Detection, Bias Correction, Weak-label Inference, Dataset-shift Mitigation
- Engineering workflow: C/C++, Linux, Workflow Automation (YAML / Shell / CMake)
- Scientific computing: ROOT / RooFit, Geant4

PROFESSIONAL EXPERIENCE

Postdoctoral Researcher | CNRS / CERN / LHCb Collaboration + GRAiNITA Collaboration | 2021-2025

- Built reusable Python/C++ workflows to analyse noisy measurement data, validate model behaviour, and support reliable downstream decisions.
- Developed calibration and bias-correction workflows to improve model reliability under sample mismatch and imperfect observation conditions.
- Built monitoring and diagnostic pipelines to detect anomalies, compare expected versus observed behaviour, and support root-cause analysis.
- Developed CatBoost-based validation pipelines with train/test diagnostics, reproducible model application, and dataset-shift mitigation through reweighting.
- Implemented fit-based inference where reliable event-level ground truth was unavailable, using control samples and structured validation.
- Worked in shared technical codebases with review discipline and reproducible delivery for decision-facing analyses.
- Turned validation findings into concrete follow-up actions, including calibration updates and a software issue later fixed downstream.

Doctoral Research | University of Chinese Academy of Sciences | 2016-2021

- Built simulation-backed statistical workflows during doctoral research, combining model fitting, calibration, and bias-aware analysis on noisy measurements.

SELECTED OUTCOMES

- Improved reconstructed-energy resolution by about 2x through time-resolved signal analysis and calibration on realistic measurement data.
- Quantified that detector-response non-uniformity contributed less than 1% to system-resolution degradation in a measurement-to-model validation study, helping separate true design risk from modelling noise.

Education: Joint Master and PhD in Particle Physics - University of Chinese Academy of Sciences, 2016-2021 | BSc in Applied Physics - China University of Mining and Technology, 2010-2014 | Languages: Chinese: Native speaker, English: C1, French: B1 to B2